PRODUCT USER MANUAL

For GLOBAL_OMI_TEMPSAL
sst_area_averaged_anomalies
sst_trend

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/08/16	All	Creation of the document	S. Good	
1.1	2020/09/10		Addition of 2019	S. Good	
1.2	2021/08/05	All	New document template and addition of 2020	S. Good	
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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Sea Surface Temperature (SST) Ocean Monitoring Indicators (OMIs) for the global ocean. The OMIs are:

- GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies, which is a time series of monthly global average SST anomalies relative to the 1991 – 2020 average. A filtered version of the monthly time series is also included.
- GLOBAL_OMI_TEMPSAL_sst_trend, which provides a global map of linear trends in SST between 1982 and 2024.

In addition, a global map is provided of SST anomalies relative to the 1991 – 2020 average.

These OMIs are derived from SST analyses produced by the Copernicus Climate Change Service (C3S) (product SST-GLO-SST-L4-REP-OBSERVATIONS-010-024).

The product is published on the CMEMS dissemination unit and is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow the CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The products are described in the associated Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies GLOBAL_OMI_TEMPSAL_sst_trend
Geographical coverage	Global
Variables	Monthly and filtered monthly anomalies (variable names: sst_anomaly and sst_anomaly_filtered) Map of linear SST trend between 1982 and 2024 (variable name: sst_trends)
Available time series	Time series: 01/1982 – to -1Y
Temporal resolution	Time series: monthly
Horizontal resolution	Map of trends: 0.05° In latitude and longitude
Integration depth (for example)	Surface
Format	NetCDF4

b) Production System Description

GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies: Global and monthly average time series between 1982 and 2024 relative to the 1991 – 2020 average, with filtered SST anomalies obtained using the X11-seasonal adjustment procedure. This OMI is derived from the C3S SST analyses (product SST-GLO-SST-L4-REP-OBSERVATIONS-010-024).

GLOBAL_OMI_TEMPSAL_sst_trend: linear trends in SST for the global ocean over the period 1982 – 2024. This OMI is derived from the C3S SST analyses (product SST-GLO-SST-L4-REP-OBSERVATIONS-010-024).

c) Dataset details

GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies contains monthly and global average SST anomaly time series (unfiltered and filtered) derived from the C3S L4 product (SST-GLO-SST-L4-REP-OBSERVATIONS-010-024). Anomalies are computed relative to the 1991-2020 reference period. The filtered anomaly is generated using the X11 seasonal adjustment method.
sst_anomaly [K] Area Averaged Monthly SST Anomalies derived from ESA SST CCI and C3S SST Analyses sea_surface_temperature **sst_anomaly_filtered [K]** Trend Component of Area Averaged Monthly SST Anomalies derived from C3S SST Analyses sea_surface_temperature
GLOBAL_OMI_TEMPSAL_sst_trend contains a 2D map of linear SST trends over the global ocean and derived from the C3S L4 product (SST-GLO-SST-L4-REP-OBSERVATIONS-010-024).
sst_trends [K] Sea surface temperature trends 1982-2024 sea_surface_temperature

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-GLO-SST.pdf